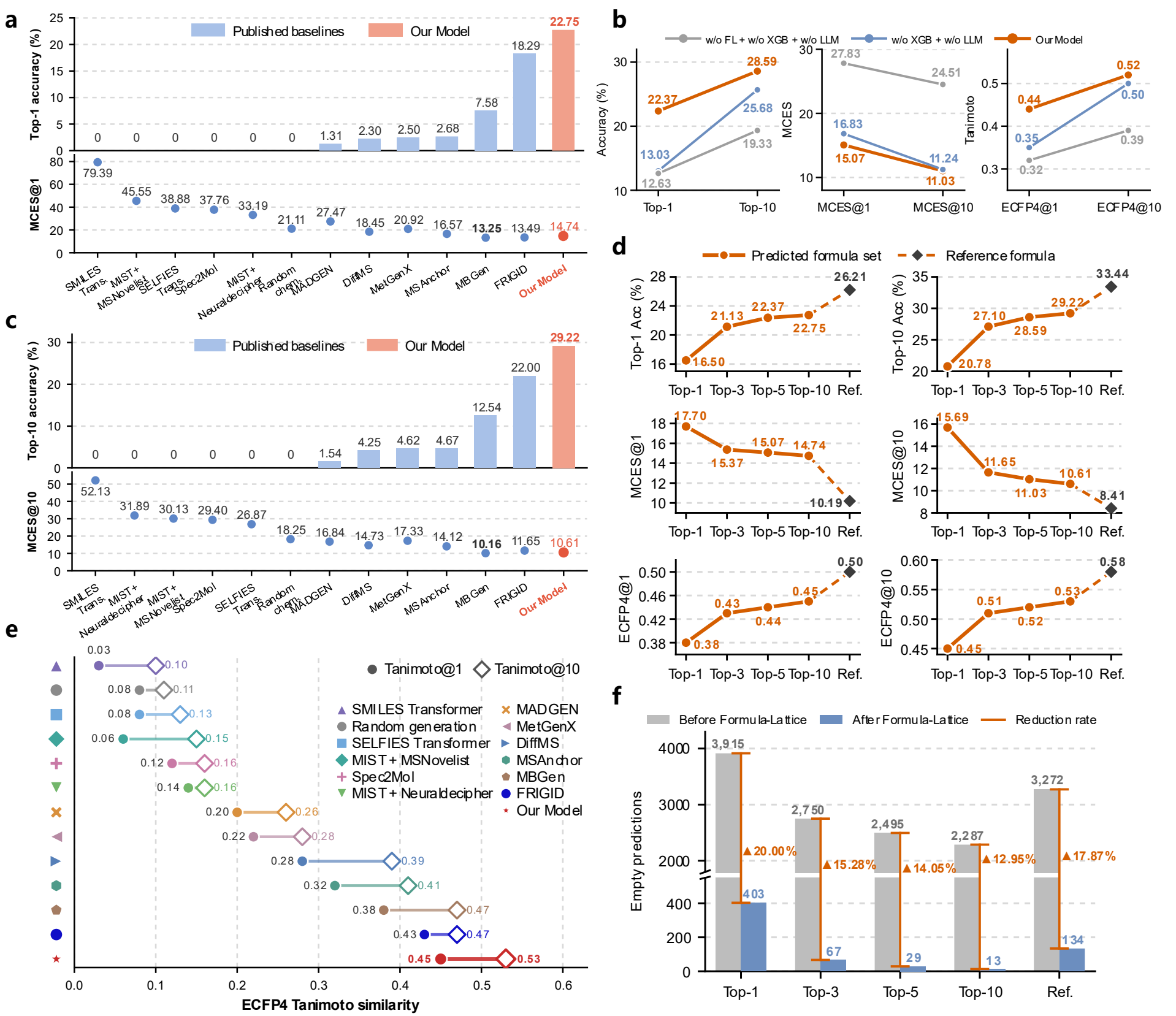

a
Published baselines
Our Model
Top-1 accuracy (%)
MCES@1
SMILES Trans.
MIST+ MSNovelist
SELFIES Trans.
Spec2Mol
MIST+ Neuraldecipher
Random chem.
MADGEN
DiffMS
MetGenX
MSAnchor
MBGen
FRIGID
Our Model
b
w/o FL + w/o XGB + w/o LLM
w/o XGB + w/o LLM
Our Model
Accuracy (%)
MCES
Tanimoto
Top-1
Top-10
MCES@1
MCES@10
ECFP4@1
ECFP4@10
c
Published baselines
Our Model
Top-10 accuracy (%)
MCES@10
d
Predicted formula set
Reference formula
Top-1 Acc (%)
Top-10 Acc (%)
MCES@1
MCES@10
ECFP4@1
ECFP4@10
Top-1 Top-3 Top-5 Top-10 Ref.
e
Tanimoto@1
Tanimoto@10
SMILES Transformer
Random generation
SELFIES Transformer
MIST + MSNovelist
Spec2Mol
MIST + Neuraldecipher
MADGEN
MetGenX
DiffMS
MSAnchor
MBGen
FRIGID
Our Model
ECFP4 Tanimoto similarity
f
Before Formula-Lattice
After Formula-Lattice
Reduction rate
Empty predictions
Top-1
Top-3
Top-5
Top-10
Ref.